

ISE 555 HW 1

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Question 1

To maximize the number of assembled components that can be produced as a linear program denote x_1 = number of shafts produced per year and x_2 = number of frames produced per year. Each assembly requires exactly one shaft and one frame, $Z = \min(x_1, x_2)$. The number of complete assemblies is maximized when $x_1 = x_2$ since any additional units would be wasted, so the objective is $\max Z = x_1$.

Maximize: $Z = x_1$

Subject to:

- $0.6x_1 + 0.8x_2 \leq 90000 = \text{Forging (20 * 4500)}$
- $0.3x_1 \leq 5 * 4500 = \text{Lathe (5 * 4500)}$
- $0.2x_2 \leq 13500 = \text{Drilling (3 * 4500)}$
- $0.3x_2 \leq 27000 = \text{Milling (6 * 4500)}$
- $0.4x_1 + 0.6x_2 \leq 45000 = \text{Grinder (10 * 4500)}$
- $x_1 = x_2$
- $x_1, x_2 \geq 0$

The optimal solution is $x_1^* = x_2^* = 45000$, giving $Z^* = 45000$ assembled units per year.

Numerical Check

Substitute $x_1 = x_2 = 45000$ into each constraint:

- Forging: $0.6(45000) + 0.8(45000) = 63000 \leq 90000$
- Lathe: $0.3(45000) = 13500 \leq 22500$
- Drilling: $0.2(45000) = 9000 \leq 13500$
- Milling: $0.3(45000) = 13500 \leq 27000$
- Grinder: $0.4(45000) + 0.6(45000) = 45000 \leq 45000$ (minimum capacity, bottleneck)

The solution was chosen from the Grinder constraint. All other constraints are satisfied.

Optimality Certificate

Setting $x_1 = x_2 = x$, the objective becomes $Z = x$, which is increasing in x . Substituting into each constraint bounds x from above:

$$x \leq 64286 \text{ (Forging)}, \quad x \leq 75000 \text{ (Lathe)}, \quad x \leq 67500 \text{ (Drilling)},$$

$$x \leq 90000 \text{ (Milling)}, \quad x \leq 45000 \text{ (Grinder)}.$$

Again the minimum capacity of these upper bounds is the Grinder at $x = 45000$, which is the optimal solution.

Divisibility

Optimal Solution is already whole number, rounding is not necessary.

Question 2

Given 3 ingredients, corn x_1 , limestone x_2 and alfalfa x_3 each with their own cost per bushel. Additionally, given the required bushel amounts of N and nutritional requirements for each ingredient written for corn would be $c_{11}, c_{12}, c_{13}, c_{14}$, limestone $c_{21}, c_{22}, c_{23}, c_{24}$ and alfalfa $c_{31}, c_{32}, c_{33}, c_{34}$ for the nutritional content of each ingredient. The nutritional requirements must result in the combined nutritional values falling between a range of $\{P_1, P_2, P_3, P_4\}$ at its lower bound and $\{u_1, u_2, u_3, u_4\}$ as its upper bound per bushel. The objective is to determine the optimal amount of ingredients to order to minimize the cost and meet the needs.

Minimize: $Z = 12x_1 + 5x_2 + 8x_3$

Subject to:

- $x_1 + x_2 + x_3 = N$ (Total Bushel Requirement)
- $P_1 * N \leq c_{11}x_1 + c_{21}x_2 + c_{31}x_3 \leq u_1 * N$ (Protein Requirement)
- $P_2 * N \leq c_{12}x_1 + c_{22}x_2 + c_{32}x_3 \leq u_2 * N$ (Calcium Requirement)
- $P_3 * N \leq c_{13}x_1 + c_{23}x_2 + c_{33}x_3 \leq u_3 * N$ (Carbohydrate Requirement)
- $P_4 * N \leq c_{14}x_1 + c_{24}x_2 + c_{34}x_3 \leq u_4 * N$ (Vitamin Requirement)
- $x_1, x_2, x_3 \geq 0$ (Non-negativity Constraint)

Problem description asks for a linear program, and due to the limited values in the problem statement, the solution is not possible to determine.

Question 4

Part(A) - Using a standard transportation/assignment model, the problem can be formulated as bidders "sources" as a_i , depot "destinations" as b_j , and the cost of shipping from each bidder to each deposit as c_{ij} . The decision variable is the amount shipped from each bidder to each deposit, denoted as x_{ij} .

Minimize: $Z = \sum_i \sum_j c_{ij} x_{ij}$

Subject to:

- $\sum_j x_{ij} \leq a_i$ (Supply constraint)
- $\sum_i x_{ij} \geq b_j$ (Demand constraint)
- $x_{ij} \geq 0$ (Non-negativity constraint)

Part(B) - Now, adding a discount to the unit delivery cost if $\sum_j x_{ij} \geq a_i$. With discounted rate now being d_{ij} , if a discount is applied at z_i the new model becomes:

Minimize: $Z = \sum_i \sum_j c_{ij} x_{ij} z_i + \sum_i \sum_j d_{ij} x_{ij} z_i$

Subject to:

- if $\sum_j x_{ij} \leq a_i$ then $z_i = 0$ else $z_i = 1$ (Discount constraint)
- $\sum_i x_{ij} \geq b_j$ (Demand constraint)
- $x_{ij} \geq 0$ (Non-negativity constraint)

Question 6

Members of the Committee, thank you for your time. It seems that in all of Operations Research work to make concise models to explain system behaviors and inform decision making they continually fail at making a concise and direct definition for me to use. I've seen it described as 'The art of winning the war without actually fighting it' which seems too concise and leaves you all with more questions than answers and much more verbose 'The application of methods of science to complex problems in the direction and management... Develop a scientific model of the system, incorporating measurements of factors such as change and risk with which to predict and compare the outcomes of alternative decisions, strategies or controls. Help management determine its policy and actions scientifically'.

While these reasoning make Operation Research seem like a political science its root is in the decision making with analytical data behind it. I don't need to tell members of congress the importance of a good baseball metaphor but Operation Research is the Moneyball decision making in finding the value where the human eye misses.

The application in operations research is in billions of dollars in industry from budgeting and investments, purchasing, marketing, defense and nearly every aspect of commerce.

I would like to ask the members of the committee is there a strategy to the Department of Energy in providing electricity to 340 million Americans is there a strategy? In supporting thousands of cities across the country is there a strategy? In this very room do you members have a strategy?

There are 2 counter arguments I see with Operations Research the first is the commitment of time and money to build out models for the slightly improved solutions worth it? The price of experimenting on the grid itself rather than Operations models exceeds any cost it may incur. The second and less applicable argument is it takes a certain feeling out of the game and causes decision makers to over rely on statistics and numbers than making the right decisions, but luckily with yourselves as management I see no reason why these additional resources can support you in making the right decisions.